

characterize dual solutions f, g

generalized notion of convexity

Why do we need these?

- In the OT dual problem, we get conditions like: $f(x) + g(y) \leq c(x, y)$.
- Need tools to **characterize dual solutions** f, g that satisfy such conditions
- As a bonus, get **generalized notion of convexity** that is determined by $c(x, y)$ and the geometry it encodes
- Motivation:
 - Just as convex analysis gives us a powerful language for optimization,
 - OT duality forces us to invent a cost-dependent geometry to express dual potentials.

c-transforms